PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	A systematic review of strategies adopted to scale up COVID-19 testing in low- middle and high-income countries
AUTHORS	Muttamba, Winters; O'Hare, Bernadette; Saxena, Vibhor; Bbuye, Mudarshiru; Tyagi, Parul; Ramsay, Andrew; Kirenga, Bruce; Sabiiti, Wilber

VERSION 1 – REVIEW

REVIEWER	Hansoti, Bhakti Johns Hopkins University School of Medicine
REVIEW RETURNED	17-Mar-2022

GENERAL COMMENTS	Overall this is in an interesting paper, likely the low number of papers identified and included is likely secondary to a lag time bias. This is an extremely well-written article and is an interesting and reflective piece. Especially as we move from "test, test and test" and "test and treat". My only major criticism is that the language throughout the results is very colloquial i.e.,- High-income countries introduced drive-through testing to bring testing closer to the people. Sample collection and referral was mainly deployed by the low income and the lower-middle-income countries, while communitywide testing using community health workers was done by the middle-income countries." Each of the sub-sections in the results can be improved with more precise scientific reporting. I would have preferred to have had the discussion to culminate into a conceptual framework for scale-up of testing and go as far to suggest a strategy for scale-up for future testing/pandemic needs.

REVIEWER	Magno, Laio Universidade do Estado da Bahia, Salvador, Brasil, Departamento de Ciências da Vida
REVIEW RETURNED	31-May-2022

GENERAL COMMENTS	I am grateful for the opportunity to review an article with high relevance for global public health. The article brings a systematic review with scope to encompass several studies and reports published over a vast period of the COVID-19 pandemic. The article may provide important insights for police makers, researchers and stake holders working with COVID-19 testing around the world. I will point out some concerns that may contribute to the improvement of the manuscript. The main issue of concern is with the methodological detailing in the search for articles and reports.

	The latter, for example, seems to me to be underrepresented in the review, especially since only reports in English were included. Generally, this type of document is made to respond to local needs. Furthermore, the google tool seems limited to finding reports from the most diverse countries on the planet. In this sense, I suggest that this review be limited to articles published in indexed journals, at the risk of generating wrong conclusions about non-English-speaking countries. Or even consider including other languages in the search for reports. Below I will point out other issues according to the parts of the article. Title: consider including "low-income" as said in the first paragraph of methods. Abstract: consider standardizing the objective of the abstract with that of the end of the introduction. Supplementary Table 1 – Consider organizing publications according to the socioeconomic level of the countries. The countries of #15 (line 57) are missing; Figure: what is the title of the Figure 1? (p. 5) Introduction: line 38 – change from “isn't” to “is not”. Methods: Provide registration information for the review, including register name and registration number, or state that the review was not registered (see the #24a PRISMA 2020 Checklist). Indicate where the review protocol can be accessed, or state that a protocol was not prepared (see the #24b PRISMA 2020 Checklist). Standardize term “articles” or “studies”; reports can be the result of studies carried out? What is the definition of “reports” reviewed in the article? (p. 6) Line 21-26: say which criterion or reference was used to define the economic level of countries. Topic “inclusion/exclusion criteria”: how studies were grouped for the syntheses? (see the PRISMA 2020 Checklist). (p. 6) Line 37-40: it is not clear how the translation of the publications in policy and practice by the countries was analyzed. This criterion is more related to an empirical study objective. How to check this translation from a review? Topic “Search strategy” It is necessary to detail how each strategy reached the articles and reports. A table with each strategy and the number of publications achieved helps the reader to verify the scope of each of the strategies used. Can be added as supplemental material. Specify the date when each source was last searched or consulted (see the #6 PRISMA 2020 Checklist);

	PICO: detail each of the topics covered for this review. That is, what are the patient/population, intervention, comparison and outcomes of this review? Please check #10 PRISMA 2020 Checklist: List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect. List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information. Please check #11 PRISMA 2020 Checklist: Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process. Please check #14 PRISMA 2020 Checklist: Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases). Topic “Data access and management” What internet pages were used? Was there duplication analysis? Topic “Database” How many researchers did the review? I found it described in the line Line 5-6 (p. 8). So, consider bringing that information up beforehand. What is the level of agreement among researchers? What was the “predefined criteria” used to assess quality? Where is this analysis presented in the article? Did articles and reports have the same criteria? (p. 7) Line 58-59: I don't understand the difference to high-income country. Topic: “data abstraction”- did you mean “data extraction”? (p. 8) Line 17-33 – Have all articles and reports been read or was a pre-screening performed based on titles and abstracts? Consider showing the codebook as supplemental material. Exemplify what was “something not captured by the codebook”. Topic “Data analysis” What themes were considered? Was country income considered in this analysis? Results Please check PRISMA 2020 Checklist: #19Present assessments of risk of bias for each included study.

	#20a For each synthesis, briefly summarize the characteristics and risk of bias among contributing studies. #22 Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed. Discussion List limitations after reviewing the topics mentioned above. Conclusion Review after reviewing the topics mentioned above.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1: Dr. Bhakti Hansoti, Johns Hopkins University School of Medicine

Overall this is in an interesting paper, likely the low number of papers identified and included is likely secondary to a lag time bias. This is an extremely well-written article and is an interesting and reflective piece. Especially as we move from "test, test and test" and "test and treat".

Response: Thank you for the kind comment.

My only major criticism is that the language throughout the results is very colloquial i.e.,- High-income countries introduced drive-through testing to bring testing closer to the people. Sample collection and referral was mainly
deployed by the low income and the lower-middle-income countries, while communitywide testing using community health workers was done by the middle-income
countries."

Response: We have worked on the language and now believe the manuscript reads better

Each of the sub-sections in the results can be improved with more precise scientific reporting. I would have preferred to have had the discussion to culminate into a conceptual framework for scale-up of testing and go as far to suggest a strategy for scale-up for future testing/pandemic needs.

Response: We have worked on this and in the second last paragraph of the discussion, we give a summary of what the review revealed and what this means for future pandemics. In summary, we make mention that most of the strategies involved using existing disease systems for test development and roll out, however, these need to be accompanied by flexible regulatory and legislative systems (Page 22).

Reviewer 2: Dr. Laio Magno, Universidade do Estado da Bahia, Salvador, Brasil, Universidade Federal da Bahia

Comments to the Author:

I am grateful for the opportunity to review an article with high relevance for global public health. The article brings a systematic review with scope to encompass several studies and reports published over a vast period of the COVID-19 pandemic. The article may provide important insights for policy makers, researchers and stake holders working with COVID-19 testing around the world.
Response: Thank you for the kind comment.

I will point out some concerns that may contribute to the improvement of the manuscript. The main issue of concern is with the methodological detailing in the search for articles and reports. The latter, for example, seems to me to be underrepresented in the review, especially since only reports in English were included. Generally, this type of document is made to respond to local needs. Furthermore, the google tool seems limited to finding reports from the most diverse countries on the planet. In this sense, I suggest that this review be limited to articles published in indexed journals, at the risk of generating wrong conclusions about non-English-speaking countries. Or even consider including other languages in the search for reports.

Response: We acknowledge articles in other languages other than English are missing. We indicated in the inclusion criteria that we limited our search to only articles in English. We acknowledge this as a limitation and have added the following statement: Further, the review only considered scholarly articles in English, and thus there might be underrepresentation of strategies used in the non-English speaking countries (Page 22).

Title: consider including "low-income" as said in the first paragraph of methods.

Response: This has been done. Title now reads: “A systematic review of strategies adopted to scale up COVID-19 testing in low, middle and high-income countries” (Page 1).

Abstract: consider standardizing the objective of the abstract with that of the end of the introduction.
Response: Thank you. This has been done.

Supplementary Table 1 – Consider organizing publications according to the socioeconomic level of the countries. The countries of #15 (line 57) are missing.

Response: The missing countries have been added. Regarding organising publications according to the socioeconomic level of the countries, this is not possible as the strategies were not exclusive i.e., some strategies were used by countries across all economic profiles).

Figure: what is the title of the Figure 1?
Response: Thank you. The title has been added ie Flow diagram of included publications.

(p. 5) Introduction: line 38 – change from “isn't” to “is not”.
Response: Thank you. This has been done (Page 20).

Methods:

Provide registration information for the review, including register name and registration number, or state that the review was not registered (see the #24a PRISMA 2020 Checklist). Indicate where the review protocol can be accessed, or state that a protocol was not prepared (see the #24b PRISMA 2020 Checklist).

Response: We have added that the review was not registered. We state as follows: This study was not registered and a full review protocol was not written as it was originally conducted as a rapid review and later changed to a full systematic review (Page 23).

Standardize term “articles” or “studies”; reports can be the result of studies carried out? What is the definition of “reports” reviewed in the article?

Response: We have changed and use the term “scholarly articles”. We go further go on to list what these articles were. The following statement has been added: The scholarly articles included original studies, situational and practice reports, case reports, features, commentaries, viewpoints, reviews and perspectives (Page 6).

(p. 6) Line 21-26: say which criterion or reference was used to define the economic level of countries.
Response: This criterion is mentioned in the section of Geo-economic coverage of included publications. We used GDP per capita in USD, and a reference for this is included. (Page 10)

Topic “inclusion/exclusion criteria”: how studies were grouped for the syntheses? (see the PRISMA 2020 Checklist).
Response: This is described under section of data analysis (Page 9).

(p. 6) Line 37-40: it is not clear how the translation of the publications in policy and practice by the countries was analyzed. This criterion is more related to an empirical study objective. How to check this translation from a review?
Response. Thank you for highlighting this. We realise the statement was misleading and have now framed it better to under inclusion/exclusion criteria and now reads: We excluded any scholarly articles that proposed strategies to scale up testing, but with no mention of whether these strategies had been implemented (Page 6).

Topic “Search strategy”
It is necessary to detail how each strategy reached the articles and reports. A table with each strategy and the number of publications achieved helps the reader to verify the scope of each of the strategies used. Can be added as supplemental material.

Response: A table summarising this has been drawn and submitted as a supplementary table 3.

Specify the date when each source was last searched or consulted (see the #6 PRISMA 2020 Checklist)
Response: The date when each source was last searched has been added. Under database, we have added that all sources were last searched on 17th September 2021 (Page 8).

PICO: detail each of the topics covered for this review. That is, what are the patient/population, intervention, comparison and outcomes of this review?
Response: This detail has been added under the search strategy. The following statement has been added: The population was the individuals in the countries, intervention was the COVID-19 testing, while the outcome was COVID-19 testing strategies countries. The study did not have a comparison (Page 7).

Please check #10 PRISMA 2020 Checklist:
List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.

Response: The section on data abstraction has been expanded to clarify on how this was done. The main outcome were the testing approaches used and compatibility of the outcome was through use of a predeveloped codebook (Pages 8-9).

List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.

Response: This has been done under section of data abstraction. The other variable sought was the COVID-19 testing methods used (Page 9).

Please check #11 PRISMA 2020 Checklist:
Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.

Response: Given the evidence synthesis method used and the kind of information sought for, assessment of the risk of bias was not possible.

Please check #14 PRISMA 2020 Checklist:
Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).

Response: This was not applicable for the kind of review we did.

Topic “Data access and management”
What internet pages were used? Was there duplication analysis?

Response: The internet sources have been added and are now listed as PubMed, Google scholar, and google search engine (Page 7).

Topic “Database”

How many researchers did the review? I found it described in the line Line 5-6 (p. 8). So, consider bringing that information up beforehand. What is the level of agreement among researchers?

Response: This has been done. We state that 3 researchers did the review. Regarding agreement, we mention that the researchers through consensus were able to agree on the final list to consider (Page 7).

What was the “predefined criteria” used to assess quality? Where is this analysis presented in the article? Did articles and reports have the same criteria?

Response: This was misleading and the statement has been deleted.

(p. 7) Line 58-59: I don't understand the difference to high-income country.

Response: The comment is not clear, but we have now used the terms low, middle and high income countries.

Topic: “data abstraction”- did you mean “data extraction”?

Response: Thank you. We meant extraction. This has been rectified (Page 8).

(p. 8) Line 17-33 – Have all articles and reports been read or was a pre-screening performed based on titles and abstracts? Consider showing the codebook as supplemental material. Exemplify what was “something not captured by the codebook”.

Response: All articles and reports were read. As advised, a codebook has been availed as supplementary material.

Topic “Data analysis”
What themes were considered? Was country income considered in this analysis?

Response: The themes have been added in the section of data analysis as: accessibility, diagnostics, regulatory and legislation, integration, personnel, technology and innovation and collaboration (Page 9).

Country income was not included in the main analysis but rather in the categorisation of strategies used by country

Results

Please check PRISMA 2020 Checklist:

#19Present assessments of risk of bias for each included study.

Response: Summary statistic for each scholarly article reviewed has been added as supplementary Table 2.

#20a For each synthesis, briefly summarize the characteristics and risk of bias among contributing studies.
Response: Summary statistic for each scholarly article reviewed has been added as supplementary Table 2.

#22 Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.

Response: We are unable to describe the extent of certainty but we do not the evidence from the review. This is added under the section of testing strategies used by the countries and in the conclusion.

Discussion

List limitations after reviewing the topics mentioned above.

Response: I have expanded the limitations section. Importantly, most of the PRISMA checklist items do not apply for the evidence synthesis methods used. We have however incorporated the checklist items to the extent possible (Page 22).

Conclusion

Review after reviewing the topics mentioned above.
Response: This has been reviewed.

VERSION 2 – REVIEW

REVIEWER	Magno, Laio Universidade do Estado da Bahia, Salvador, Brasil, Departamento de Ciências da Vida
REVIEW RETURNED	05-Sep-2022

GENERAL COMMENTS	Thank you for considering my suggestions. I would like to inform you that the topic "Data extraction" is written as "Data abstraction" and a language review is needed before the final publication. For example, contraction in the abstract and study limitation ("didn't").